

Meso2D: User Tutorial

Practical guide for new users

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Contents

1	Purpose of this tutorial	2
2	What the program does	2
3	Prerequisites	2
3.1	Dependencies	2
3.2	Run commands	3
4	Quick workflow (5 steps)	3
5	Start screen: mode selection	3
6	Project configuration	3
6.1	Main parameters	3
6.2	Gradation table	4
7	Simulation options	4
7.1	Pores	4
7.2	Reactive points	4
8	Running the simulation	4
8.1	Before pressing Run	4
8.2	During execution	5
8.3	After execution	5
9	Visualization and image saving	5
10	Structure export	5
10.1	Available formats	5
10.2	Best practices	5
11	Project lifecycle actions	5
12	Reproducibility and traceability	6
13	Troubleshooting	6
13.1	Simulation does not start	6
13.2	Too many placement failures	6
13.3	Export fails	6

14 Recommended usage habits	6
15 Quick start example	7
16 Summary	7

1 Purpose of this tutorial

This guide explains how to use Meso2D to generate 2D mesostructures with:

- aggregates (circular, elliptical, or polygonal),
- pores,
- reactive points,
- and exportable outputs for post-processing.

The approach is practical: follow a clear sequence to run reproducible simulations and export consistent data.

2 What the program does

Meso2D builds a 2D specimen and fills it with geometric entities according to:

- gradation/dosification,
- aggregate area fraction,
- pore area fraction,
- reactive-point area fraction,
- and random seed control.

The execution is orchestrated by a simulation controller that selects a worker for:

- **circulos**,
- **ellipses**,
- **poligonos**.

3 Prerequisites

3.1 Dependencies

Make sure at least the following packages are installed:

- Python 3.x
- PyQt5
- numpy
- matplotlib
- ezdx
- shapely
- opensimplex

3.2 Run commands

From the project root:

```
python Meso2D_refactor.py
```

For the English-oriented branch:

```
python Meso2D_refactor_EN.py
```

4 Quick workflow (5 steps)

1. Choose the geometry mode (circles, ellipses, or polygons).
2. Set specimen size and gradation parameters.
3. Optionally enable pores and/or reactive points.
4. Run the simulation and inspect visualization.
5. Export structure and image.

5 Start screen: mode selection

At startup, the mode selection window determines:

- aggregate placement algorithm,
- geometric representation of results,
- collision-check strategy.

Recommendation:

- Start with *circulos* for fast checks.
- Use *ellipses* or *poligonos* for more realistic morphology studies.

6 Project configuration

6.1 Main parameters

Parameter	Meaning and effect
x, y	Specimen dimensions (width and height). Total area is $A = x \cdot y$.
Pagg	Target area fraction occupied by aggregates.
Gradation table sieve_size / tpp	Defines the gradation curve. Area is split by fraction using these columns.
dporo_min, dporo_max	Minimum and maximum pore diameters.
Pporos	Target pore area fraction.
dpto_min_aggregates, dpto_max_aggregates	Diameter range for reactive points on aggregates.

Ppto_react_aggregates	Reactive-point area fraction over aggregates.
dpto_min_pasta,	Diameter range for reactive points in paste.
dpto_max_pasta	
Ppto_react_pasta	Reactive-point area fraction in paste.
seed	Random seed for exact reproducibility.

6.2 Gradation table

- Use **Add** to insert rows.
- Fill **sieve_size** and **tpp** consistently.
- Physically consistent values improve convergence and placement success.

7 Simulation options

7.1 Pores

When pores are enabled:

- pore radii are generated until target pore area is consumed,
- pores are placed without overlap against aggregates or existing pores.

7.2 Reactive points

Reactive points can be generated in two channels:

- **on aggregates**,
- **in paste**.

Internally:

- aggregate channel typically uses **cluster sampling**,
- paste channel typically uses **Simplex noise gating**.

8 Running the simulation

8.1 Before pressing Run

Recommended checklist:

1. Correct mode selected.
2. Valid x and y values.
3. Complete gradation table.
4. Pagg set.
5. If pores/reactive points are enabled, all required fields are filled.
6. Fixed seed if strict reproducibility is needed.

8.2 During execution

- Progress bar advances by pipeline stages.
- Information panel shows worker status messages.
- **Stop** can cancel the current run.

8.3 After execution

After a successful run:

- structure visualization is available,
- export actions are enabled,
- simulation time is logged.

9 Visualization and image saving

- **View structure**: opens the viewer window.
- **Save Image**: saves PNG/JPEG/TIFF/BMP.

10 Structure export

10.1 Available formats

From **Export Structure**, you can save to:

- DXF
- SVG
- JSON
- Gmsh GEO

10.2 Best practices

- Export right after a successful run.
- If parameters change, rerun before exporting.
- Include date and seed in file names for traceability.

11 Project lifecycle actions

- **New Project**: clears current state.
- **Open Project**: loads a saved project and repopulates UI fields.
- **Save Project**: stores current parameters.
- **Load Example**: loads a sample project for quick testing.

12 Reproducibility and traceability

To reproduce exactly the same structure:

1. save the project,
2. record the seed,
3. keep the same mode,
4. run the same code version.

13 Troubleshooting

13.1 Simulation does not start

Common causes:

- missing required field,
- incomplete gradation table,
- non-numeric input in numeric fields.

13.2 Too many placement failures

- reduce target fractions (Pagg, Pporos, reactive ratios),
- increase specimen size (x, y),
- reduce minimum diameters,
- begin with circles mode for calibration.

13.3 Export fails

- ensure a fresh structure has been generated,
- check write permissions in destination folder,
- test DXF export first as a baseline.

14 Recommended usage habits

1. Keep project versions and seeds documented.
2. Change one parameter group at a time.
3. Save both image and structure for every useful run.
4. Record mode, seed, Pagg, Pporos, and reactive ratios.
5. For batch studies, calibrate in circles mode and then switch to ellipses/polygons.

15 Quick start example

1. Run `python Meso2D_refactor_EN.py`.
2. Select *circulos*.
3. Set `x=100`, `y=100` (example), `Pagg`, and gradation table values.
4. Enable pores with moderate values.
5. Press **Run**.
6. Visualize and export to DXF + PNG.

16 Summary

Meso2D combines gradation-driven area budgeting, stochastic sampling, and geometric collision constraints to build 2D mesostructures.

A robust routine is:

1. configure parameters,
2. run,
3. validate visually,
4. export,
5. and record seed plus project file.

This process provides reproducible and comparable outputs for later analysis.